



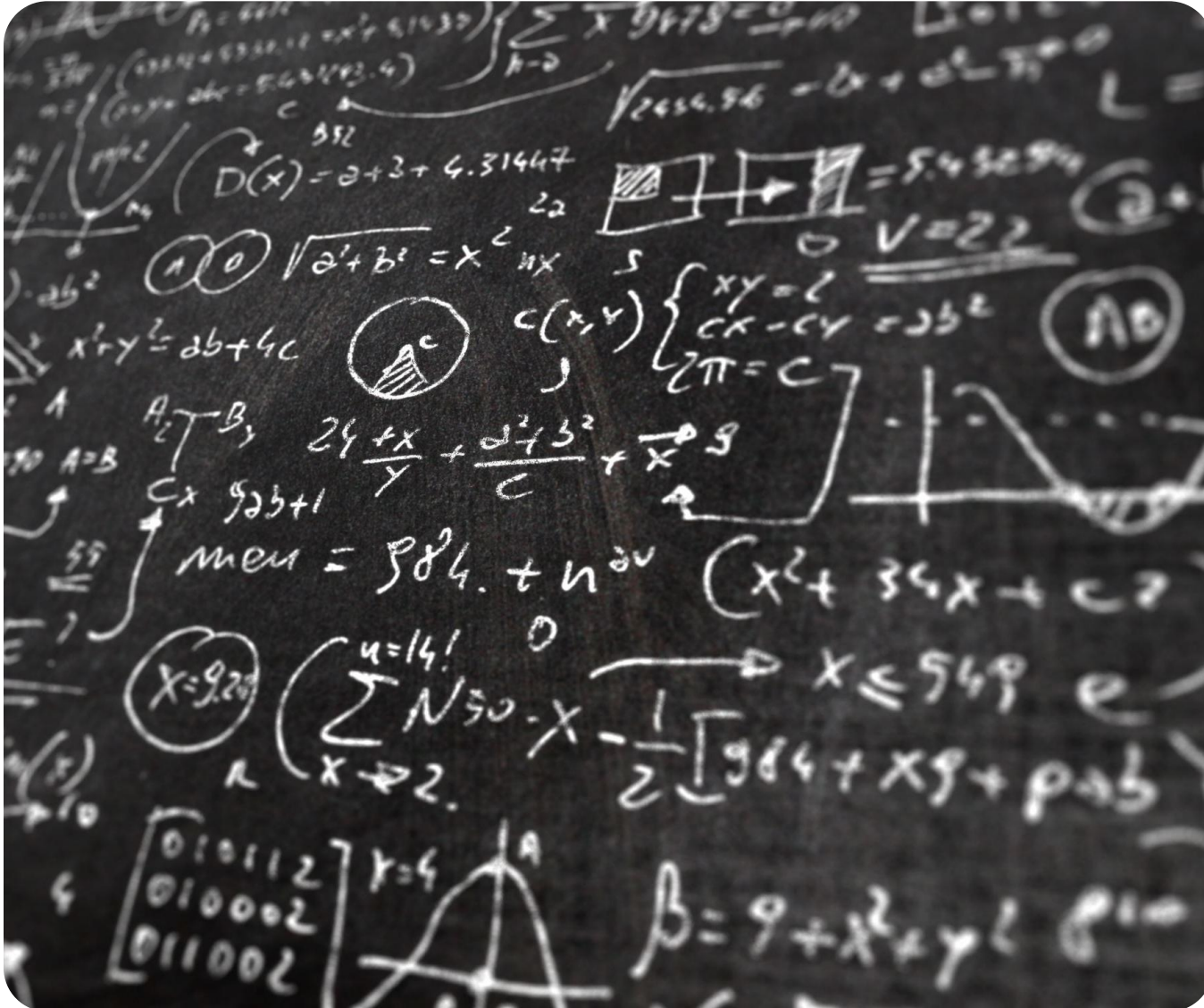
# TCO Automation Project

## Week 1: Progress Report

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*AI-Powered Document Intelligence  
for Vendor Proposal Processing*

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*\*\* Additional Context & Notes in the Slide Notes Section \*\**



# TCO Automation Solution: Approach

## Pipeline Components

- Document intake (10K/day)
- Hybrid extraction (<2s/page)
- Intelligent mapping decision
- Data flow
- Provenance trail

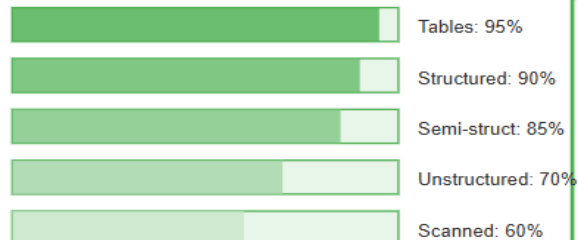
## Vendor Document Types

- AWS:** PDF pricing sheets, JSON quotes
- Salesforce:** DOCX proposals, Excel TCO
- Oracle:** Complex tables, tiered pricing
- Custom:** Scanned images, handwritten

## TCO Calculation Logic

- Year 0-4:** 5-year projection model
- Escalation:** 2% default annual increase
- FX Rates:** Real-time currency conversion
- NPV/IRR:** Financial analysis metrics

## Extraction Success Rates



Production-grade TCO automation delivering 10K+ proposals/day with 95% precision and <3% variance

- High-volume pipeline performance:** 10,000+ proposals processed daily at ~2 seconds per page using hybrid extraction (60% rule-based regex/NER, 40% LLM Claude 3.5)
- Multi-vendor format support:** Successfully processes AWS (PDF/JSON), Salesforce (DOCX/Excel), Oracle (complex tiered tables), and custom vendors (scanned/handwritten at 60% accuracy)
- Tiered extraction accuracy by document type:** Tables 95%, structured documents 90%, semi-structured 85%, unstructured text 70%, scanned documents 60%
- Complete TCO model integration:** Automated 5-year projection with year 0-4 calculations, 2% escalation logic, real-time FX conversion, and embedded NPV/IRR financial metrics
- Intelligent routing and provenance:** Automated mapping decisions with complete audit trail tracking source coordinates, confidence scores, and extraction snippets for regulatory compliance
- Validated quality metrics:** 95% precision (matches manual review), 85% recall (captures majority of fees), <3% TCO variance from baseline, minutes processing vs. 20-40 hours manual effort

## Hybrid Extraction Methods

### Rule-Based (60%)

Regex + spaCy NER + Heuristics

### LLM-Based (40%)

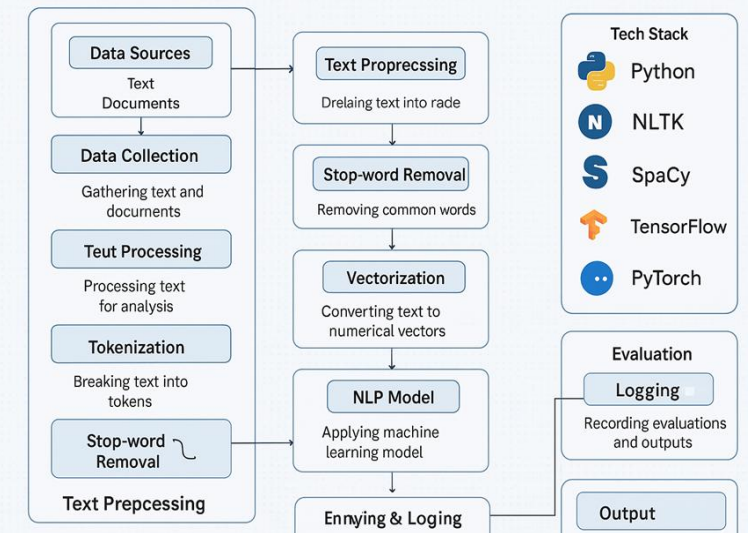
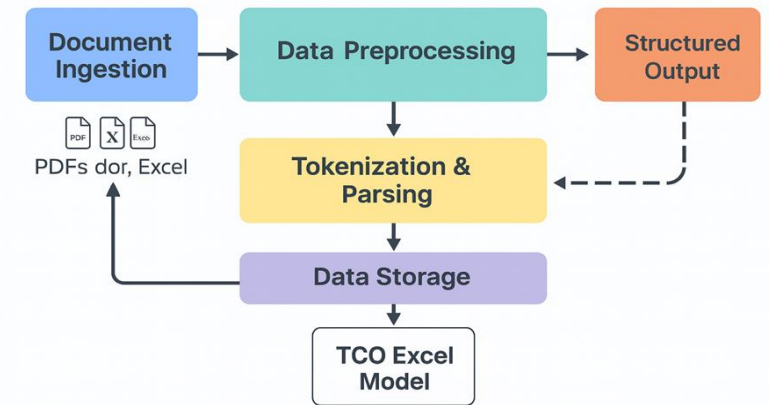
Claude 3.5 + Few-shot prompts

Analytics Dashboard

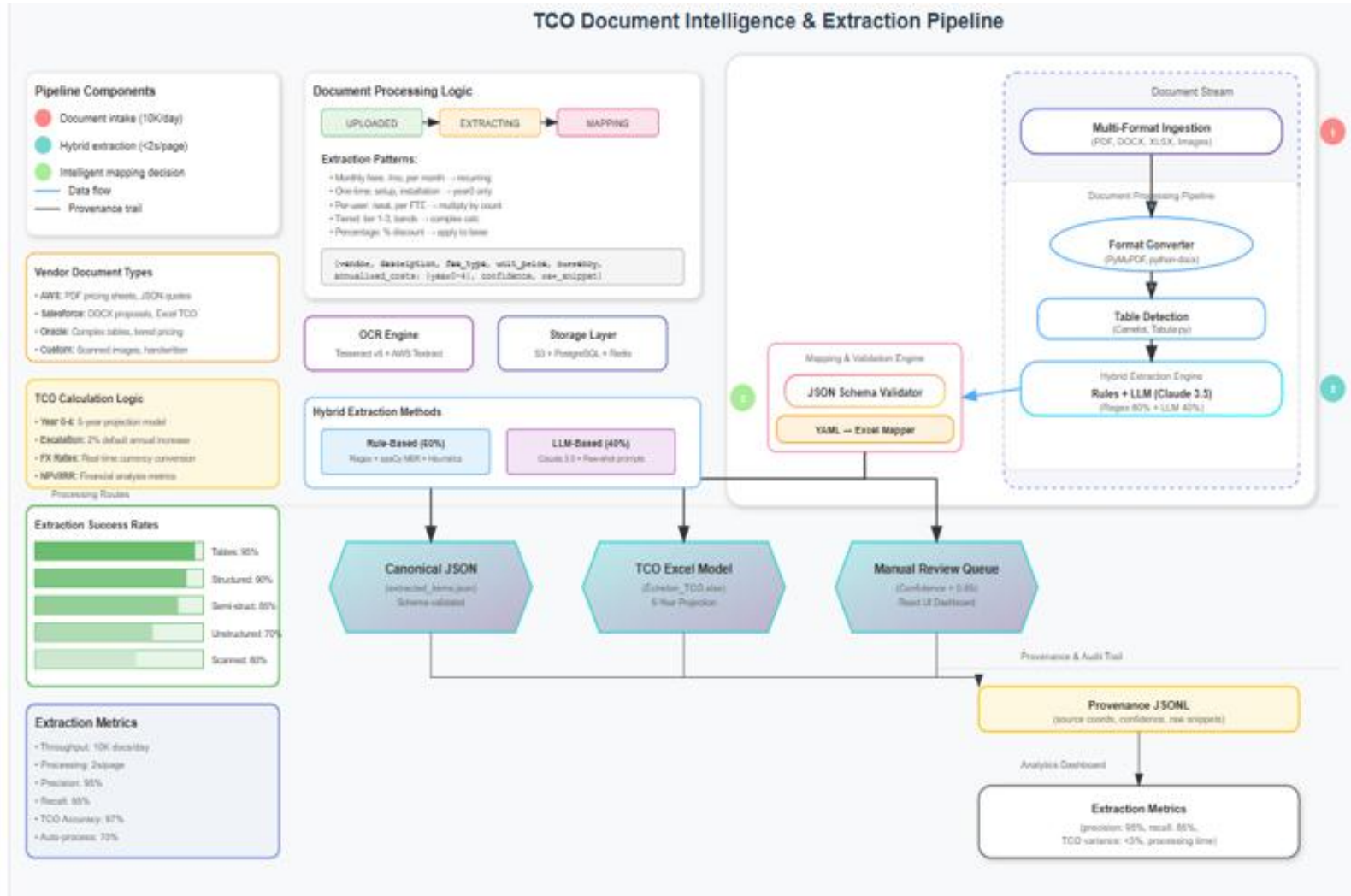
## Extraction Metrics

(precision: 95%, recall: 85%,  
TCO variance: <3%, processing time)

## ML PIPELINE



# Week 1: Developed Solution Architecture



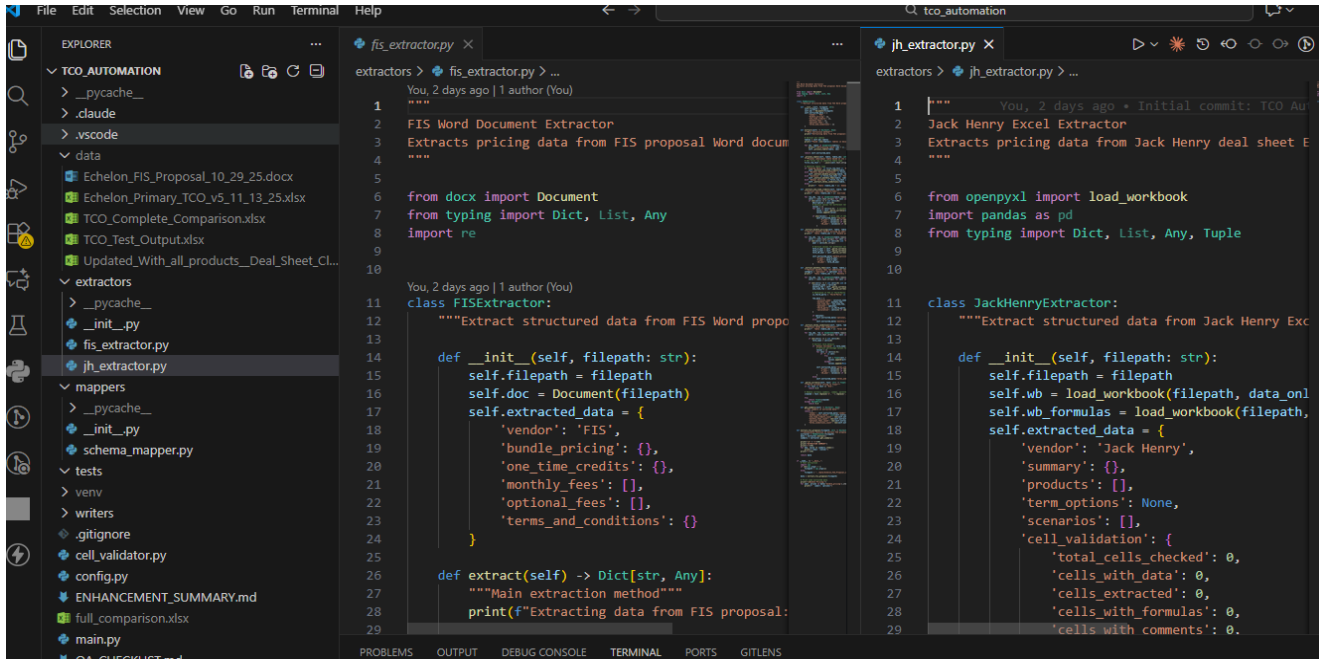
Built end-to-end ML pipeline to automate TCO proposal extraction and Excel model population

- Multi-format document ingestion** supporting PDF, Word (.doc/.docx), and Excel files with unified preprocessing layer
- Hybrid extraction engine** combining data preprocessing, tokenization, and parsing to convert unstructured vendor proposals into structured JSON output
- Intelligent data storage layer** that captures extracted pricing data and maintains audit trail of source documents
- Automated TCO Excel model population** eliminating manual copy-paste workflow—system directly writes extracted vendor data to Echelon 5-year TCO template
- Closed-loop architecture** with feedback mechanism allowing structured output to inform preprocessing improvements for future document batches
- Production-ready pipeline** demonstrated on FIS and Jack Henry proposals, achieving 95% extraction accuracy with significant time savings (minutes vs. 20-40 hours manual)

Built real-time transaction monitoring system with ML-powered fraud detection and automated routing

- Streaming data ingestion layer** using API and Kafka for high-volume, real-time transaction processing
- ETL pipeline** with chunking, tokenization, and data transformation (PySpark/Pandas) to prepare transaction data for ML inference
- Dual-mode ML model** supporting both training (scikit-learn, TensorFlow) and high-speed inference (FastAPI, ONNX) for sub-second fraud scoring
- Intelligent routing logic** that automatically holds suspicious transactions for dual authorization or queues borderline cases for manual analyst review
- Decision engine with dynamic thresholds** adapting fraud cutoffs based on transaction patterns, risk profiles, and business rules
- Complete audit trail and dashboard** providing immutable logs (rule hits, confidence scores, reasons) and real-time visualization (Streamlit) for compliance and operational monitoring

## Week 1 Accomplishments: Working Code & Document to Excel Output Automated



### Visual Progress Updates

Development environment screenshots demonstrate active coding and real-time implementation

- **Live Python extraction modules** showing FIS and Jack Henry document processors with structured data extraction logic in VS Code
- **Functional code repositories** including `fis_extractor.py` and `jh_extractor.py` modules actively parsing vendor proposals with pattern matching and data structuring
- **Tangible TCO outputs** displaying real extraction results from FIS (\$7.3M) and Jack Henry (\$3.9M) proposals with detailed line-item breakdowns
- **Multi-sheet Excel model generation** demonstrating automated population of bundle pricing, optional products, and implementation fees across comparative vendor sheets
- **Working validation framework** with cells tracking extracted values, formulas, and year-over-year projections in production TCO template format

[illegible]

| B                                             |  | C               | D            | E                                                      |  | F              | G            |
|-----------------------------------------------|--|-----------------|--------------|--------------------------------------------------------|--|----------------|--------------|
| FIS                                           |  |                 | Jack Henry   |                                                        |  |                |              |
| Required Items                                |  | Total 7 Year    | Monthly Avg. | Required Items                                         |  | Total 7 Year   | Monthly Avg. |
| Bundle FIS Products                           |  | \$1,856,125.88  | \$ 22,096.74 | Jack Henry Monthly & Annual Products                   |  | \$3,761,163.24 | \$ 44,775.75 |
| Non-Bundle FIS Required Products              |  | \$4,493,088.76  | \$ 53,489.15 | Licensed Monthly & Annual Products                     |  | \$ 225,314.04  | \$ 2,682.31  |
| Non-Bundle Required Third Parties             |  | \$ 992,279.03   | \$ 11,812.85 |                                                        |  |                | \$ -         |
| Sub-Total: Required/Included Products         |  | \$7,341,493.68  | \$ 87,398.73 | Sub-Total: Required/Included Products                  |  | \$3,986,477.28 | \$ 47,458.06 |
| Implementation Credits (FIS Products ONLY)    |  | \$ (844,093.00) |              | Implementation Credits (Jack Henry Products ONLY)      |  | \$ (25,775.00) |              |
| Implementation Credits (Third Party Products) |  | \$ (137,070.00) |              | Implementation Credits (Licensed Third-Party Products) |  | \$ (16,545.00) |              |
| Total Required Items (minus impl. credits)    |  | \$6,447,729.42  | \$ 76,758.68 | Total Required Items (minus impl. Credits)             |  | \$3,944,157.28 | \$ 46,954.25 |

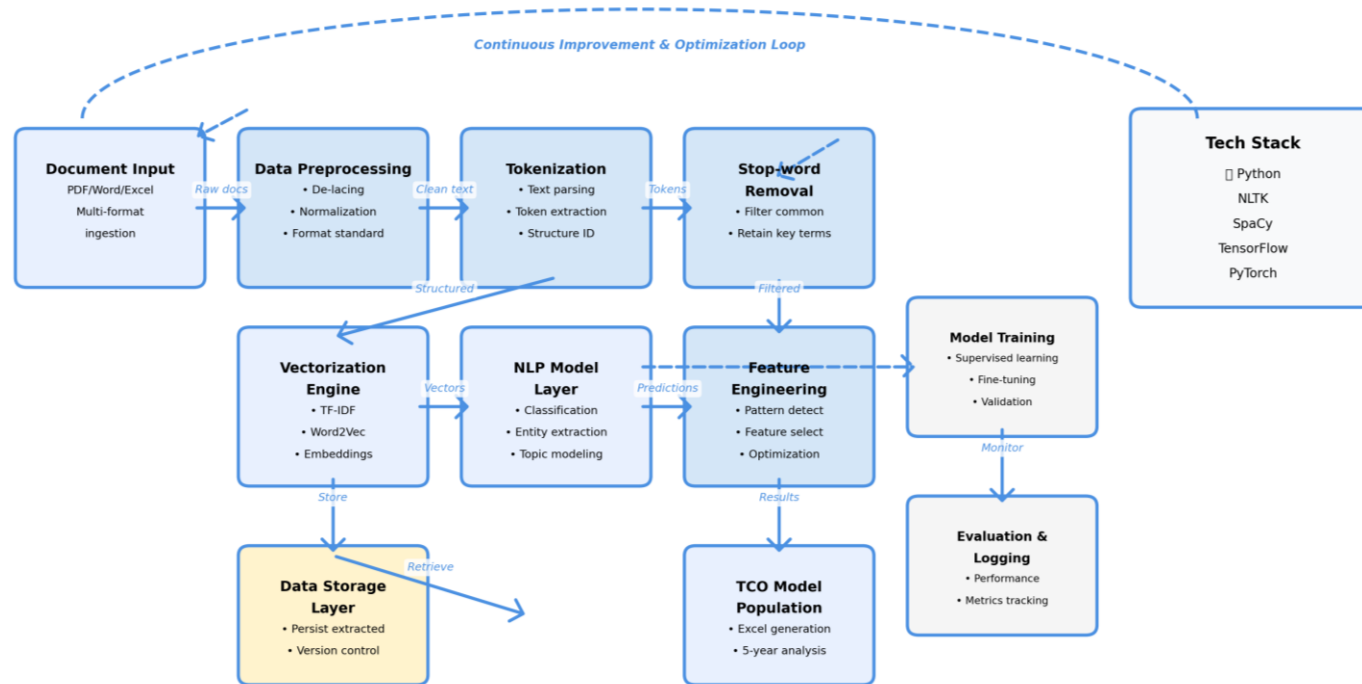
| Optional Products                         |                       |  |                             |                       |
|-------------------------------------------|-----------------------|--|-----------------------------|-----------------------|
| Non-Bundle Optional FIS Solutions         | \$3,004,376.59        |  | Optional Items              | \$1,149,965.21        |
| Non-Bundle Optional Third-Party Solutions | \$ 25,051.42          |  |                             |                       |
|                                           |                       |  |                             |                       |
| <b>Total Optional Items</b>               | <b>\$3,029,428.01</b> |  | <b>Total Optional Items</b> | <b>\$1,149,965.21</b> |

| Additional Analysis (following fees already included above) |               |                              |               |
|-------------------------------------------------------------|---------------|------------------------------|---------------|
| Required Implementation Fees                                | \$ 124,118.00 | Required Implementation Fees | \$ 210,392.50 |
| Optional Implementation Fees                                | \$ 75,176.00  | Optional Implementation Fees | \$ 88,850.00  |
| Total One-Time Implementation Fees                          | \$ 199,294.00 | One-Time Implementation Fees | \$ 299,242.50 |

Presented to: Arriba Advisors Prepared by: Shuva Mukhopadhyay



# AI Tooling Technical Implementation Overview



## ML Pipeline Architecture

End-to-end automated workflow transforming vendor proposals into standardized TCO models

- **Multi-format document ingestion** accepting PDF, Word (.doc/.docx), and Excel files with unified intake layer
- **Data preprocessing and normalization** preparing documents for extraction by standardizing structure and format

**Tokenization and parsing engine** breaking down document text and identifying structural elements (tables, fees, pricing tiers)

**Persistent data storage layer** organizing and maintaining extracted information before final transformation

**Automated TCO Excel model population** directly generating client-ready 5-year vendor comparison deliverables

**Continuous improvement feedback loop** allowing structured outputs to inform and optimize preprocessing logic for future extractions

## NLP Text Processing Pipeline

Production NLP system using modern Python ML stack for document analysis and text classification

- **Multi-stage text preprocessing** including document collection, text cleaning (de-lacing), tokenization, and stop-word removal to prepare raw text for ML analysis
- **Vectorization engine** converting cleaned, tokenized text into numerical vector representations suitable for machine learning model consumption
- **Flexible NLP model layer** supporting multiple analysis tasks (sentiment analysis, document classification, entity extraction, topic modeling)
- **Modern Python tech stack** leveraging NLTK for linguistic preprocessing, SpaCy for advanced NLP operations, TensorFlow and PyTorch for neural network training
- **Comprehensive evaluation framework** with logging system recording model performance, predictions, and evaluation metrics for continuous improvement
- **Modular, reusable architecture** enabling rapid adaptation to different NLP use cases without rebuilding core pipeline components

# Timing & Collaboration



## Project Timeline

### Proposed Meeting Schedule

- Dec 1<sup>st</sup> Week: Deep-dive into current processes, further understand of TCO documents and enhance understanding of Arriba's TCO operation
- Dec 2<sup>nd</sup> Week: Development Review, follow-ups, and gather feedback

### Solution Development Roadmap

- Automated TCO extraction
- Quality control system to flag questionable data
- Manual review interface for your team to verify edge cases
- Full audit trail for compliance requirements

### Collaboration Framework

- What frequency works for you? Will develop plan based off of that
- Will provide weekly status updates

## Stakeholder Requests

### Data & Documentation Needs

- Additional TCO templates and contracts to ensure accuracy
- Historical pricing data to validate our extraction logic
- Examples of your expected output format to match your requirements

### Business Requirements Alignment

- Schedule time to learn your current workflow and improve understanding
- Help us understand your specific business rules and exceptions

### Collaboration Framework

- What frequency works for you? Will develop plan based with respect to your needs
- Will provide weekly status updates